



Grade:

30

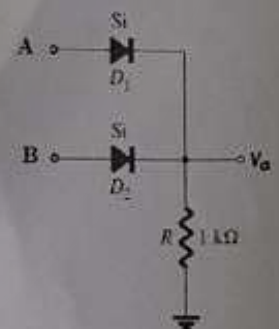
Student Name:

Student ID:

Instructor Name:

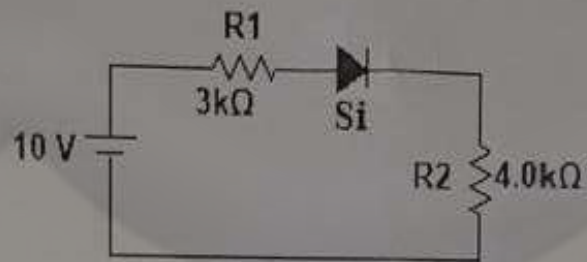
Q1. Answer the following questions from Q1 – Q20:

1. The semiconductor device has temperature coefficient according to the of conductivity with the application of heat.
a. Negative, decreasing. b. Positive, decreasing. c. Negative, increasing. d. Positive, increasing.
2. When design circuit with Zener diode, when the Zener is removed from the circuit to determine its equivalent circuit of it and then $V = V_{\text{open-circuit}}$ is greater than or equal to V_Z , the Zener diode is
a. ON and V_Z is used. b. OFF and open circuit is used. c. ON and V is used. d. OFF and V is used.
3. When a diode characteristic reaches to the breakdown region, the diode like a Zener diode is
a. Working as a switch. b. Destroyed. c. Working as a voltage regulator. d. Going in OFF state.
4. By applying a negative bias to a diode, the depletion region is and the diode current is
a. Increased, increased. b. Decreased, increased. c. Increased, decreased. d. Decreased, decreased.
5. The output frequency of half-wave rectifier is given as
a. $f_{\text{out}} = f_{\text{in}} / 2$ b. $f_{\text{out}} = 2 * f_{\text{in}}$ c. $f_{\text{out}} = f_{\text{in}} / 4$ d. $f_{\text{out}} = f_{\text{in}}$
6. The equivalent circuit for a diode in piece-wise linear model contains
a. V_{Knee} and ideal diode. b. V_{Knee} , R_{Static} , and ideal diode. c. V_{Knee} , R_{Average} , and ideal diode. d. Ideal diode.
7. During the bias switching of a diode, the required period for the minority carriers to return to their majority-carrier state in the opposite material is called....
a. Storage time. b. Transition time. c. Reverse recovery time. d. Travel time.
8. The V_{DC} value of a full-wave bridge rectifier for peak input value V_m is given as.....
a. $V_m / 2$ b. $\pi * V_m$ c. $2 * V_m / \pi$ d. V_m / π
9. For different values of A and B of 10 V and 0 V, the given circuit below represents.....
a. OR Gate. b. NOR gate. c. NAND Gate. d. AND Gate.
10. To get a better filtering in a half-wave rectifier circuit with capacitor-input filter, the ripple factor of the circuit should be by the load resistance.
a. Decreasing, decreasing. b. Increasing, decreasing. c. Increasing, increasing. d. Decreasing, increasing.



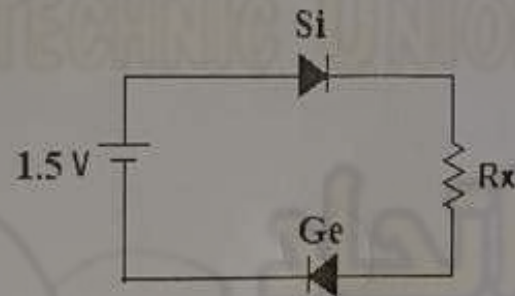
11. Assuming a simplified model is used for the diode of the circuit shown below, the voltage across the R_2 resistor is.....

- a. 0 V.
- b. 5.32 V.
- c. 1.33 mV.
- d. 9.3 V.



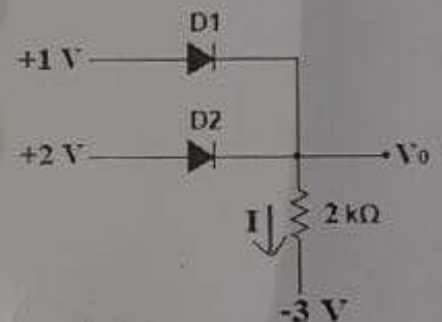
12. Assuming a simplified model is used for the two diodes of the circuit shown below and the total current of the circuit is 4 mA. Then, resistor (R_x) value is

- a. Approximately infinity.
- b. 250 Ω .
- c. 125 Ω .
- d. Can not be determined.



13. To find the current I in the following circuit assuming that both diodes are ideal, the assumption for $D1$ and $D2$ should be (R.B is Reverse bias and F.B is Forward bias).

- a. $D1$ is F.B and $D2$ is F.B.
- b. $D1$ is F.B and $D2$ is R.B.
- c. $D1$ is R.B and $D2$ is F.B.
- d. $D1$ is F.B and $D2$ is F.B.

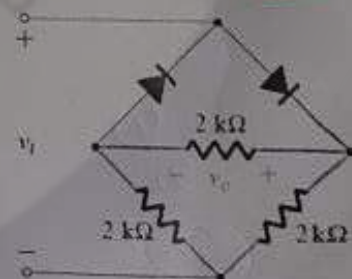
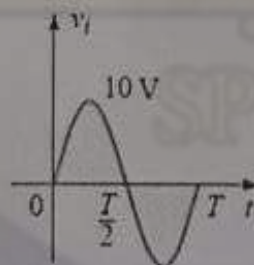


14. Based on your assumption on Q13, the value of I is

- a. 0.25 mA
- b. 2 mA
- c. 0.5 mA
- d. 2.5 mA.

15. The output voltage v_o on the middle resistor $2\text{ k}\Omega$ during the positive half cycle is..... (Assume ideal diodes).

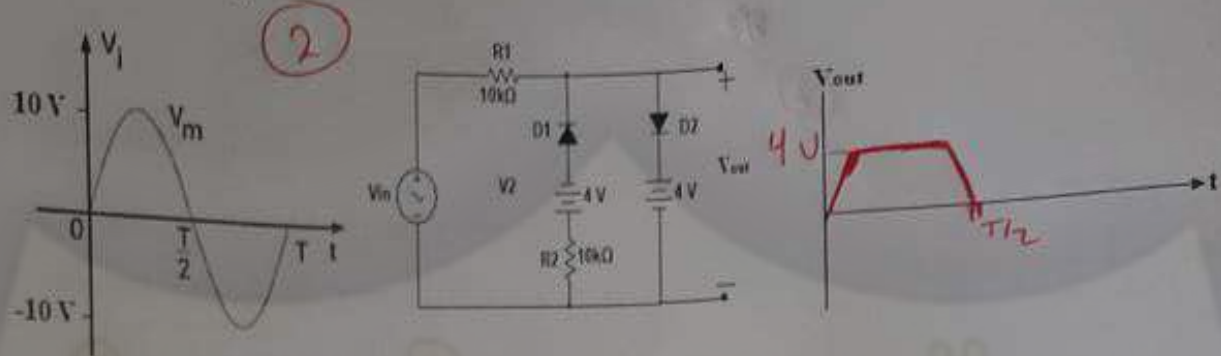
- a. 3.18 V
- b. 4.2 V
- c. 5 V
- d. 6.4 V



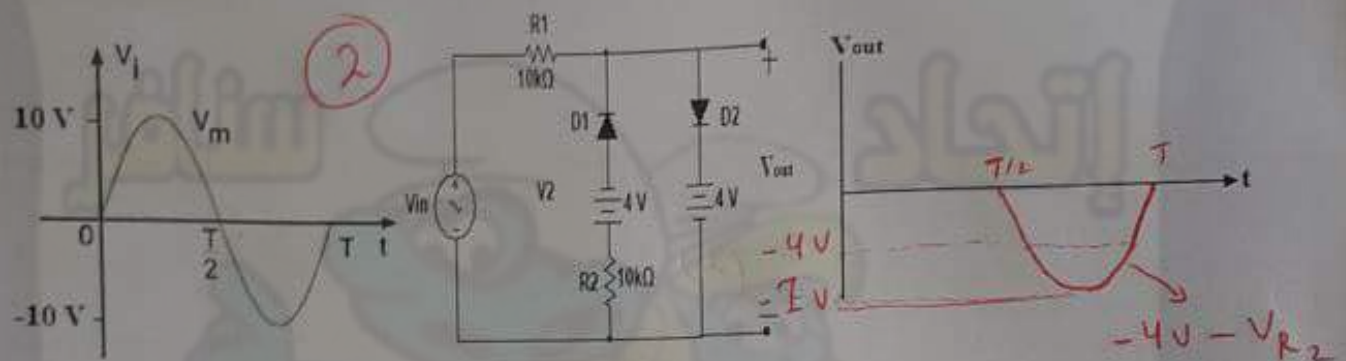
16. The PIV for the given circuit in Q15 with the same input voltage and ideal diodes is given as

- a. 0 V
- b. 0.636 V
- c. 3.18 V
- d. 5 V

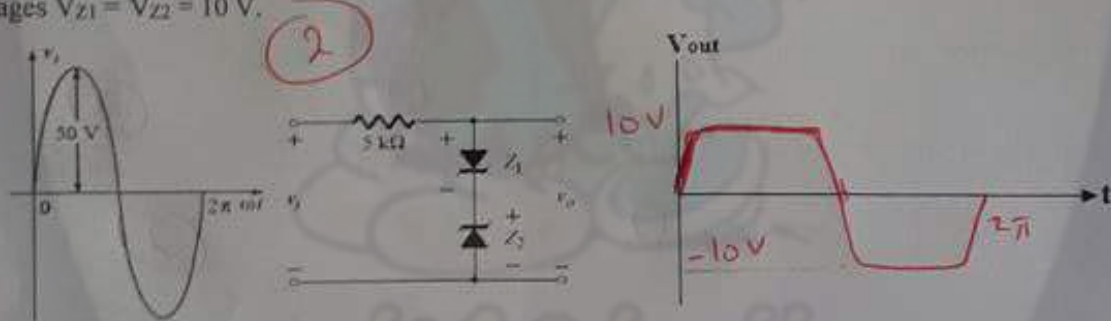
17. Draw the output voltage during the positive cycle of the following circuit based on the given input voltage. (Both diodes are ideal).



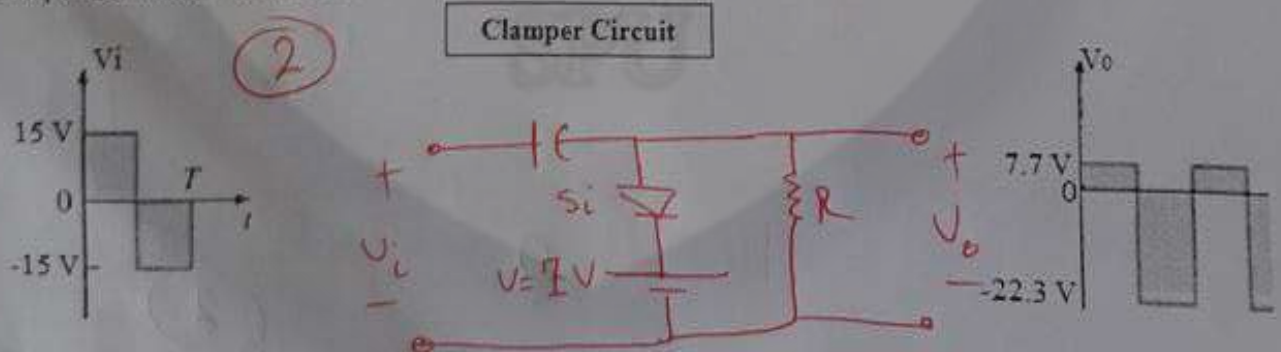
18. Draw the output voltage during the negative cycle of the following circuit based on the given input voltage. (Both diodes are ideal).



19. Draw the output voltage V_0 for the given circuit below assuming that the two diodes are ideal and the voltages $V_{Z1} = V_{Z2} = 10V$.



20. Use the given input waveform to design a clamper circuit to get the output waveform. Assume that the diode is simplified model silicon diode.



Good Luck!

Al-Balqa Applied University

Faculty of Engineering

Electrical Power Engineering Department

Grade: 30

Midterm exam of Electronics I

2nd Semester 2021-2022

April 10, 2022 10:00 am - 10:50 am

Student Name: محمود أمين عبد الكريم ياسين 12:30-11 ر - ن

Student ID: 32115333018 د. رانيا محمد، الكساسبة

Q1. Fill the following table by the correct answers of the following questions. (20 points)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
<u>d</u>	<u>d</u>	<u>b</u>	<u>d</u>	<u>c</u>	<u>a</u>	<u>d</u>	<u>b</u>	<u>c</u>	<u>a</u>

**1. When a pentavalent impurity is added to a pure semiconductor, then the semi-conductor becomes....
a. An insulator b. An intrinsic semiconductor c. Positive type semi-conductor d. Negative type semi-conductor

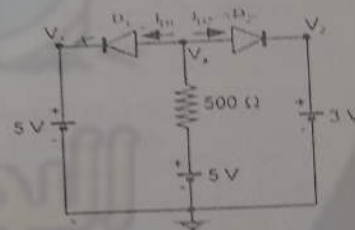
**2. What are the common reasons of diode failure?
a. Excessive forward current b. Reverse breakdown voltage
c. Power dissipation d. Both a & b

**3. In which rectifier the center-tapped transformer is required?
a. Half-wave rectifier b. Full-wave rectifier c. Filter rectifier d. All of the above

**4. Using a ripple factor of 3.9 % and $V_{oc} = 15.3$ V for a full-wave rectifier with a frequency of 120 Hz and load resistance of 220Ω , what is the best capacitor value that is used to design a filter with lower ripple.
a. $850 \mu F$ b. $976 \mu F$ c. $1000 \mu F$ d. $2500 \mu F$

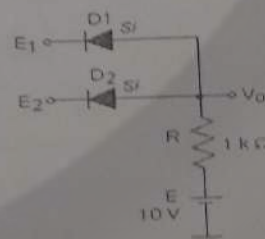
**5. Determine the values of I_{D1} and V_{D2} for the flowing circuit assuming that a simplified model of Silicon diode is used for each diode.

- a. 10 mA, 0.7 V
b. 10 mA, 0 V
c. 0 A, 0.7 V
d. Not mentioned.

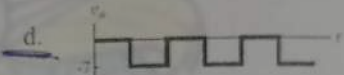
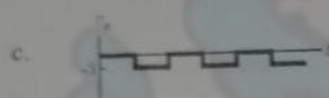
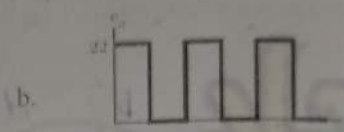
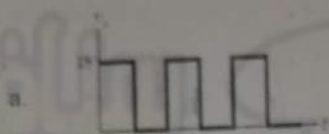
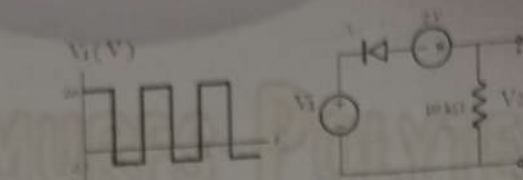


**6. If the value of $E_1 = 0$ V and $E_2 = 10$ V, what the is value of the output voltage V_0 ?

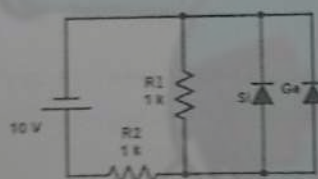
- a. 10 V b. -10 V c. 0 V d. 9.3 V.



** 7. What is output waveform for the following circuit using the given input voltage?



** 8. Using the circuit shown below, what is the current value that passes through R2?



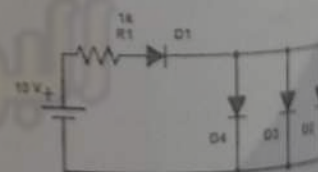
a. 9.3 mA

b. 9.7 mA

c. 10 mA

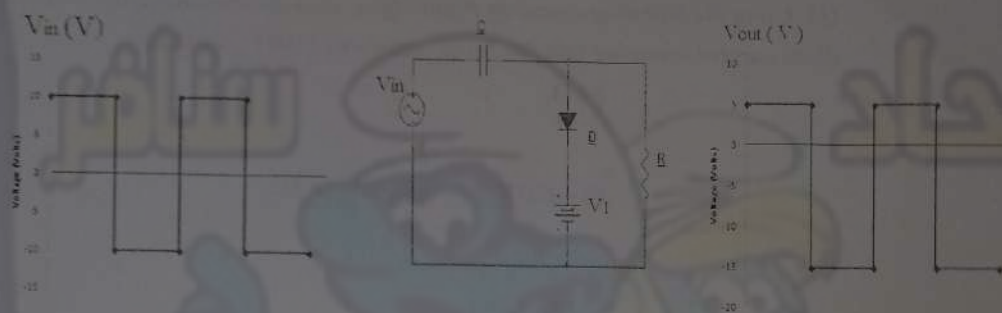
d. 0 mA

** 9. In the given circuit below, if the diode D2 is a Germanium diode and all other diodes are Silicon diodes, then which of the following statements is true?



a. $I_{D2} < I_{D3}; I_{D2} = I_{D4}$ b. $I_{D2} > I_{D3}; I_{D2} > I_{D4}$ c. $I_{D2} > I_{D3}; I_{D2} = I_{D4}$ d. $I_{D2} > I_{D3}; I_{D2} < I_{D4}$

10. Given the input and the output of the following circuit with $f = 1 \text{ KHz}$ and simplified model of Silicon diode is used, what is the value of the bias voltage V_1 ?



- a. 4.3 V b. -4.3 V c. 4V d. 5 V

Q2. Assuming a simplified model of diode is used in the circuit of Figure Q2 below, find the values the output voltage V_o and the current I_x . (5 points)

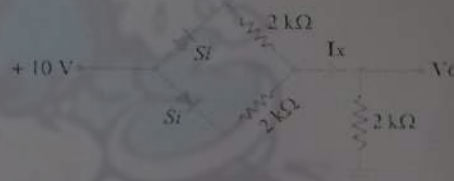


Figure Q2

$$V_o = V_m - 2V_T$$

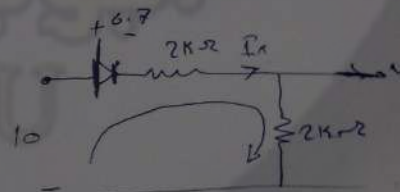
$$V_o = 10 - 2(0.7) = 8.6 \text{ V}$$

$$-10 + 0.7 + (2\text{k})I + (2\text{k})I = 0$$

$$-9.3 + 4\text{k}I = 0$$

$$4\text{k}I = 9.3$$

$$I = \frac{9.3}{4\text{k}} = 2.325 \text{ mA}$$



Q3. Using the following circuit in Figure Q3 to find the maximum and minimum values of Zener diode current assuming that the Zener voltage $V_Z = 50\text{ V}$. (5 points)

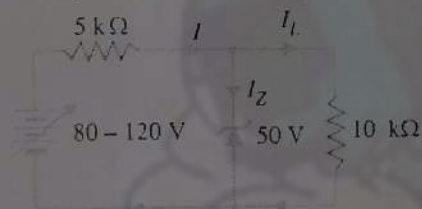


Figure Q3